

SAMPLE REVIEWER IN FILIPINO
MGA SAGISAG PANULAT NG ILANG MANUNULAT

1. Francisco Baltazar-Balagtas
2. Jose dela Cruz-Huseng Sisiw
3. Jose Corazon de Jesus-Huseng Batute, Daniel Viterbo, Luksang Paruparo, Anastacio Salagubang, Pusong Hapis
4. Iñigo Ed Regalado-Odalager, Dapit-hapon
5. Valeriano H. Peña-Kintin Kulirat at Tandang Anong; Obra-maestra: “Nena at Neneng”
6. Florentino Collantes-Kuntil Butil
7. Severino Reyes-Lola Basyang, Pedro Penduko, Silvio Ruiz, “Don Binoy”
8. Jesus Balmori-Batikuling
9. Jose P. Rizal-Laon-Laon, Dimas-Alang
10. Marcelo H. del Pilar-Pupdoh, Dolores Manapat, Piping; Dilat, Plaridel
11. Carolina Flores Trinidad-Lina Flor
12. Julian C. Balmaceda-Alphahol o Hulyo, Abril
13. Andres Bonifacio-Magdiwang, Agap-ito Bagumbayan; Kauna-unahang nagsalin sa Tagalog ng Mi Ultimo Adios
13. Emilio Jacinto -Pingkian, Dimas-ilaw
14. Emilio Aguinaldo-Magdalo
15. Graciano Lopez Jaena-Bolivar, Diego Laura
16. Amado V. Hernandez-Julio Abril, Amante Ernani
17. Lope K. Santos-Lakandalita
18. Hermogenes Ilagan-Ka Muhing Pinakatanyag na dula: “Dalagang Bukid”
19. Mariano Ponce-Tikbalang, Nanding at Kapulaki

MGA TAGURI SA ILANG MANUNULAT NG PILIPINAS

1. Manuel L. Quezon- Ama ng Wikang Pambansa
2. Lope K. Santos- Ama ng Pambansang Balarila; Apo ng mga Mananagalog; Obra Maestra: “Banaag at Sikat”; Nobela: “Salawahang Pag-ibig” - Kauna-unahang nobelang nalathala sa Pilipinas
3. Andres Bonifacio- Ama ng Katipunan; Ama ng Demokrasyang Pilipino
4. Emilio Aguinaldo- Utak ng Katipunan
5. Apolinario Mabini - Utak ng Himagsikan
6. Padre Modesto de Castro-Ama ng Tuluyang Klasika sa Tagalog; Urbana at Feliza
7. Tomas Pinpin-Kauna-unahang Manlilimbag
8. Jose Corazon de Jesus-Unang Hari ng Balagtasan; Kalipunan ng mga tula – Mga Gintong Dahon at Sa Dakong Silangan
9. Severino Reyes-Ama ng Dulaang Tagalog; Obra-maestra:“Walang Sugat”
10. Amado V. Hernandez-Ama ng Manggagawa
11. Julian Felipe-Ama ng Pambansang Awit (himig)
12. Jose Palma-Lumikha ng liriko ng Pambansang awit
12. Aurelio Tolentino-Unang Mandudulang Makabayan; Obra-maestra:“Kahapon, Ngayon at Bukas”
13. Deogracias A. Rosario-Ama ng Maikling Kwentong Tagalog; Obra-maestra: “Walang Panginoon”
14. Honorata dela Rama-Hernandez - Reyna ng Kundiman, Mutya ng Dulaang Tagalog; Primadong Tanghalang Pilipino; Reyna ng Sarswelang Tagalog
15. Francisco Baltazar-Ama ng Panulaang Tagalog
16. Claro Caluya-Prinsipe ng mga Makatang Ilokano
17. Edirberto Gumban - Ama ng Panitikang Bisaya
18. Dr. E. Arsenio Manuel-Ama ng Foklorikong Filipino
19. Pari Dr. Mariano Pilapil-Cicero ng Wikang Tagalog
20. Pedro Bukaneg-Ama ng Panitikang Ilokano
21. Valeriano H. Peña-Ama ng Nobelang Tagalog; Obra Maestra – Nena at Neneng
22. Jose Nepomuceno-Ama ng Pelikulang Pilipino
23. Fernando Monleon-Ama ng Batutian
24. Alejandro Abadilla -Ama ng Modernistang Panulaang Filipino
25. Pascual Poblete -Ama ng Pahayagan sa Pilipinas
26. Gemiliano Pineda-Ama ng Sanaysay

AWTOR AT AKDA

- Pedro Paterno
- sumulat ng nobelang panlipunan – NINAY – orihinal na wika – Kastila (pangkaugalian o pangkaasalan)
- Pascual Poblete
- naglimbag- El Gruto de Pueblo
- Padre Pedro San Buenaventura
- sumulat ng kauna-unahang talasalitaan sa Tagalog
- Graciano Lopez Haena
- nagtatag ng La Solidaridad
 - sumulat ng Fray Botod
- Marcelo H. del Pilar
- nagtatag ng Diaryong Tagalog
- Faustino Aguilar
- sumulat ng nobelang Pinaglahuan at Lihim ng Isang Pulo
- Amado V. Hernandez
- sumulat ng Tula:Isang Dipang Langit (tula); Aklasan (Tula)
 - Nobela:Ibong mandaragit
- Kauna-unahang aklat na nalimbag sa Pilipinas
- Doctrina Cristiana
- Pangalawang aklat na nalimbag sa Pilipinas
- Nuestra Señora del Rosario

Geraldine C. Rebamonte, M.A.Ed.
Cebu Normal University
Peb 2, 2013

SAMPLE REVIEWER IN EDUCATIONAL TECHNOLOGY

Competencies in Educational Technology

- Determining Appropriate Instructional Approaches and Methods, Learning Activities, Instructional Materials and Tools
- Choose appropriate principles in the preparation and utilization of the conventional and non-conventional technology tools as well as traditional and alternative teaching strategies.

Why Use Visual Aids?

- They save time and add interest
- They improve listener understanding
- They improve listener retention
- They allow for different learning styles

The Purpose of Using Visual Aids

... AVs

- support your ideas
- improve audience comprehension
- add variety to presentation (a break from listening)
- enliven a difficult / boring subject
- help illustrate complex ideas/concepts
- help the presenter to stick to the plan

Types of Visual Aids:

- Overhead Transparencies
- Slides
- Flip Charts and Posters
- Objects and Models
- Handouts
- Pocket charts
- Diorama
- Whiteboards and Chalkboards
- Computer Assisted Audiovisuals

Guidelines for VAs

- The VA should never become the REASON for the speech
- The VA should be appropriate for the length and format of the speech
- Know what technology will be available to you
- The higher the tech, the more you need redundancy and backup--Burnett’s rule of VA

Additional Guidelines for Text

- Use only 4-6 lines of text per VA
- Use phrases, not full sentences
- Use Upper Case and Lower Case for ease of reading
- Leave the same space at the top of a VA
- Limit lines to no more than 40 characters

How To . . . Conduct a Field Trip

- Planning - purpose, preview, make arrangements, arrange transportation, identify rules, provide supervision
- Preparing - build interest, provide instructions
- Conducting - arrive promptly, encourage students, obtain materials, count students
- Follow-Up - activities, thank you

Visuals

- Non projected Visuals
 - Still Pictures
 - Drawings
 - Charts
 - Graphs
 - Posters
 - Cartoons
- Projected Visuals
 - Integration

What is a graphic organizer?

- A powerful visual picture of information that allows the mind to see undiscovered patterns and relationships.
- A tool that allows organization of these patterns and relationships in writing.

Keys to effective use of graphic organizers

1. Consistent
 - Create a standard set of graphic organizers.
 - Establish a routine for implementing them in the classroom.
2. Coherent
 - Provide clear labels for the relationship between concepts in graphic organizers.
 - Limit the number of ideas covered.
3. Creative use during all stages of lesson design.
 - Incorporate during homework and test review.
 - Add illustrations.
 - Implement with cooperative groups and pairs.

Clustering → A nonlinear activity that generates ideas, images and feelings around a stimulus word. As students cluster, their thoughts tumble out, enlarging their word bank for writing and enabling them to

see patterns in their ideas.

Fishbone Mapping→ Used to show causal interaction of a complex event (an election, a nuclear explosion) or complex phenomenon (juvenile delinquency, learning disabilities).

Chain of Events→ Used to describe the stages of an event, the actions of a character or the steps in a procedure.

Interaction Outline→ Used to show the nature of an interaction between persons or groups, such as the interaction between European settlers and American Indians.

Problem/Solution→ requires students to identify a problem and consider multiple solutions and possible results

Spider Map→ Used to describe a central idea; a thing, a process, a concept, a proposition. The map may be used to organize ideas or brainstorm ideas for a writing project.

Story Board →A graphic and sequential depiction of a narrative. Students recall major events of the story then illustrate the events in the squares provided.

Review on ICT basic concepts

- An electronic device, operating under the control of instructions stored in its own memory unit, that can accept data (input), manipulate the data according to specified rules (process), produce information (output) from the processing, and store the results for future use.

History of Computers

- Charles Babbage - father of computer
 - 1800's planned analytical engine
- ENIAC - developed at end of WW II
- 1951 - 1963 1st and 2nd generation
 - very large, used unreliable vacuum tubes
- 1963 - present - 3rd and 4th generation
 - smaller, faster - use transistors and integrated circuits

Input Devices→ Any hardware component that allows you to enter data, programs, commands, and user responses into a computer. Examples: Keyboard, Mouse

Output Devices → Output devices make the information resulting from processing available for use. Examples: Printers (Impact, Nonimpact, Photo); Display Devices (CRT, LCD)

Communications Devices

- A communications device is a hardware component that enables a computer to send (transmit) and receive data, instructions, and information to and from one or more computers
- Communications occur over transmission media, such as telephone lines, cables, cellular radio networks, and satellites

Types of Computers

- Supercomputers and Mainframes
- Microcomputers
 - Desktops
 - Laptops (Notebooks)
 - Handhelds

COMPUTER SOFTWARE

- System software consists of programs to control the operations of computer equipment
- Instructions in the operating system tell the computer how to perform the functions of loading, storing, and executing an application program and how to transfer data
- When a computer is turned on, the operating system is loaded into the computer's memory from auxiliary storage, a process called booting
- Most computers use an operating system that has a graphical user interface (GUI)

Application Software

- Word Processing
- Spreadsheet
- Presentation Graphics
- Database
- Contact Management
- Utility Software – anti-virus, data backup, and file compression

Software - Legal Issues

- Commercial software
 - Can only make backup copies for yourself
 - Can only use on one machine at a time
 - Site license - use on more that one machine
- Shareware
 - Can use - make copies and give to anyone
 - Should pay if you continue to use
- Freeware – can copy and use indefinitely

Software Viruses

- Illegal code added to a program
- May spread to many computers
 - Copy files from one computer to another
 - Download files by modem
 - E-mail attachments
- Virus may be relatively harmless

- Writes "You've been stoned" on screen
- Virus may also be very damaging
 - Erases everything on hard drive
- Virus may activate on a certain date

Virus Protection

- Be careful where you copy files from
- Do not open e-mail attachments unless you are sure that it is safe
- Use virus protection program
 - Detects and removes illegal code
 - Should be updated often

Hardware vs. Software

- Hardware
 - The computer equipment
 - Includes printers, monitors, disk drives, etc.
- Software
 - Programs which tell the computer what to do
 - Examples - word processing, gradebook, tutorials, games, etc.

NETWORKS

- A network is a collection of computers and devices connected together so users can share data and resources
- A local area network (LAN) connects computers in a limited geographic area
- A wide area network (WAN) covers a large geographical area

The World Wide Web

- One of the more popular segments of the Internet is the World Wide Web, also called the Web.
- A Web page is a document that contains text, graphics, sound, and/or video and has built-in connections, or hyperlinks to other Web documents
- A Web site is a related collection of Web pages
- You access and view Web pages using a software program called a Web browser
- A Web page has a unique address, called a Uniform Resource Locator (URL)
- A hyperlink is a virtual connection to another place on the Web

LABEL:

http:// → Protocol
www. → Site
yahoo.com → Domain

ICT and Education

ICTs can be used in education to:

- improve administrative efficiency
- disseminate teaching and learning materials to teachers and students
- improve the ICT skills of teachers and students
- allow teachers and students access to sources of information from around the world
- share ideas on education and learning

Technology's Role in Restructuring Education

- Teachers will retain a key role
- Interdisciplinary approach will flourish
- Research and problem solving skills will gain attention
- Assessment methods will change to reflect the new curriculum

Some Trends Regarding Technology

- Organizational decentralization, both geographically and managerially
- Increases productivity, decreases in employees
- Dramatic increases in information sharing
- Dramatic decreases in middle level management
- Increased use of contract workers

Technology enables the following:

- Students to become motivated, engaged and learn actively
- Authentic learning
- Collaborative learning
- Explore, discover and produce knowledge
- Meets the multiple learning styles and intelligences students have
- Students to become proficiency in technology literacy skills
- Performance-based assessments

Educational Technology & Technology Education

Educational Technology (Information Technology) →Teaches with technology (uses technology as a tool) →Primarily concerned with the narrow spectrum of information and communication technologies →Primary goal: To enhance the teaching and learning process	Technology Education (Technological Studies) →Teaches about technology as a content area →Concerned with the broad spectrum of technology (How humans have designed & innovated the natural world) →Primary goal: Technological literacy for everyone
--	---

Maria Montessori → Students are lifelong learners and problem solvers
 Learning occurs in an inquiring and cooperative atmosphere
 Students increase their own knowledge through self-and teacher-initiated experiences.
 Learning takes place through the senses.
 Students learn through manipulating materials and interacting with others.
 These meaningful experiences are precursors to the abstract understanding of ideas.

Jean Piaget → "Constructivism is a philosophy of learning founded on the premise that, by reflecting on our experiences, we construct our own understanding of the world we live in. Each of us generates our own "rules" and "mental models," which we use to make sense of our experiences. Learning, therefore, is simply the process of adjusting our mental models to accommodate new experiences" (Funderstanding, 2008).

Lev Vytgosky → Children make meaning through community places, which plays a central role, and the people around the student greatly affect the way he or she sees the world.
 The type and quality of tools used for cognitive development (culture, language, important adults to the student) determine the pattern and rate of development.
The Zone of Proximal Development - problem solving skills of tasks can be placed into three categories: Those performed independently by the learner. Those that cannot be performed even with help. Those that fall between the two extremes, the tasks that can be performed with help from others.

Theories in Educational Technology

1. Anchored Instruction Theory maintains that real world contexts imbedded in educational media serve as anchors of learning in the sense that learners construct knowledge while solving the subsequent problems.

- a form of "situated" learning
- also related to problem-based learning
- Uses multimedia, web-media, or other interactive technology used to tell stories

Examples:

- internet commerce class

2. Cognitive Flexibility Theory focuses on the learning in complex and ill-structured domains. The theory focuses on the spontaneity of learning situations. It maintains that learning should be in context, however information should be presented from multiple perspectives and various case studies should be utilized in instruction. Rand Spiro is the leading theorist.
3. Constructivism argues that learning is an active process. Learners construct their own knowledge by selecting and transforming information, constructing and refining hypotheses, and decision-making. Discovery learning is one of the major themes of constructivist theory.
4. The Diffusion of Innovations Theory attempts to explain the patterns of adoption of innovations in general. Educators' adoption of technology is an important application of the principles of this theory. Rogers is the most prominent researcher of this theory

Original Theorists

- Katz (1957) :
 - media - *opinion leaders* - opinion followers
- Everett M. Rogers
 - Diffusion of Innovations (1962-95)*
 - the process by which an innovation is communicated through certain channels over time among the members of a social system

Rogers’ (1995) Diffusion of Innovation
Stages of adoption:

Awareness - the individual is exposed to the innovation but lacks complete information about it
Interest - the individual becomes interested in the new idea and seeks additional information about it
Evaluation - individual mentally applies the innovation to his present and anticipated future situation, and then decides whether or not to try it

Trial - the individual makes full use of the innovation
Adoption - the individual decides to continue the full use of the innovation
5. The Theory of Distributed Cognition is closely related to Social Constructivism in the argument it makes that cognition is not within the individual but rather it is distributed over other people and tools. The use of telecommunications technologies in education has to rely highly on distributed cognition. Major researchers in the field are Pea, Salomon, Perkins, Cole, G. Hutchins, and Norman
6. Dual coding theory

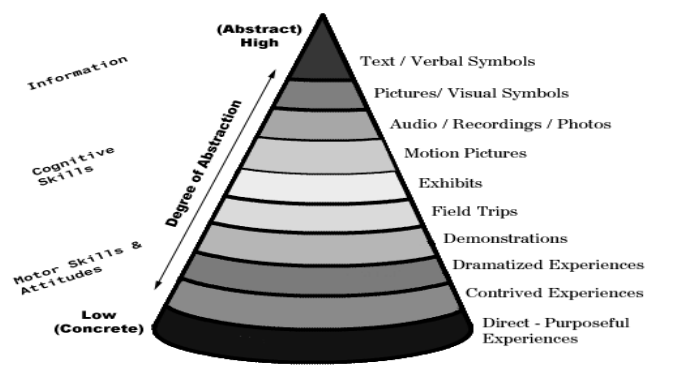
Applying Dual-Coding

- Provide a visual PowerPoint Presentation to present facts to students that include visual pictures along with written text information
- When giving students instructions on how to use a particular software try to include snapshots from the program to help students learn how to use the software
- When developing assignments for students try to give them a visual picture as an example to explain activities for students.

7. Elaboration Theory argues that instruction should be organized from simple to complex to increase learning, while providing the learner with a meaningful context in which subsequent ideas can be integrated. According to elaboration theory, instruction should be organized in increasing order of complexity for optimal learning.
8. Experiential learning addresses the needs and wants of the learner. Carl Rogers' theory is rooted in humanistic education. He identifies the two types of learning as cognitive and experiential.
9. Multiple Intelligences. The theory of multiple intelligences suggests that every individual possesses a unique combination of distinct intelligences.
10. Script Theory is primarily intended to explain language processing and higher thinking skills
11. Situated learning is a general theory of knowledge acquisition. It has been applied in the context of technology-based learning activities for schools that focus on problem-solving skills (Jean Lave & Barbara Rogoff)
12. Social constructivism maintains that learners construct knowledge in interaction with their social environment. Learning is regarded as a collaborative activity. Lev Vygotsky, is the most prominent figure in the development of social constructivist theory.
13. The symbol systems theory developed by Salomon is intended to explain the effects of media on learning. Salomon (1977) states: "To summarize, the symbol systems of media affect the acquisition of knowledge in a number of ways.

THE CONE OF EXPERIENCE

The Cone of Experience is a visual model, a pictorial device that presents bands of experience arranged according to degree of abstraction and not degree of difficulty
 This model incorporates several theories related to instructional design and learning processes



Verbal Symbols

- principal medium of communication
- bear no physical resemblance to the objects or ideas for which they stand
- may be a word for concretion, idea, scientific principle, formula or philosophic aphorism
- Disadvantage: highly abstract

Visual Symbols

- chalkboard/whiteboard, flat maps, diagrams, charts
- fits the tempo of presentation of idea, topic or situation
- very easy to procure and prepare
- Limitations: lack of ability to use the media
 - size of visuals
 - simplification of visual materials
 - leads to misconceptions

Recordings, Radio, Still Pictures

- attention – getting, particularly projected a – vies
- concretize verbal abstraction
- Limitations: size of pictures or illustrations
- expensiveness of projected

- materials and equipment
- timing difficulties between radio
- shows and classroom lessons

Exhibits

- present objects or processes otherwise impossible inside the classroom
- exposure to new ideas, discoveries, inventions
- problems that may be encountered:
 - too little space
 - time – consuming
 - maintenance

Demonstrations

- visualized explanation of an important fact or idea or process
- may require nothing more than observation or students may be asked to do what has just been shown how to do
- Disadvantages:
 - ideas or processes might not be interpreted or conceived very well
 - visibility to all learners

Field Trips

- undertaken primarily for the purpose of experiencing something that cannot be encountered within the classroom
- a rich experience in learning about objects, systems, and situations
- Disadvantages: time-consuming
 - expensive
 - high exposure to danger / accidents
 - inadequacy of the community's resources

Dramatized Experiences

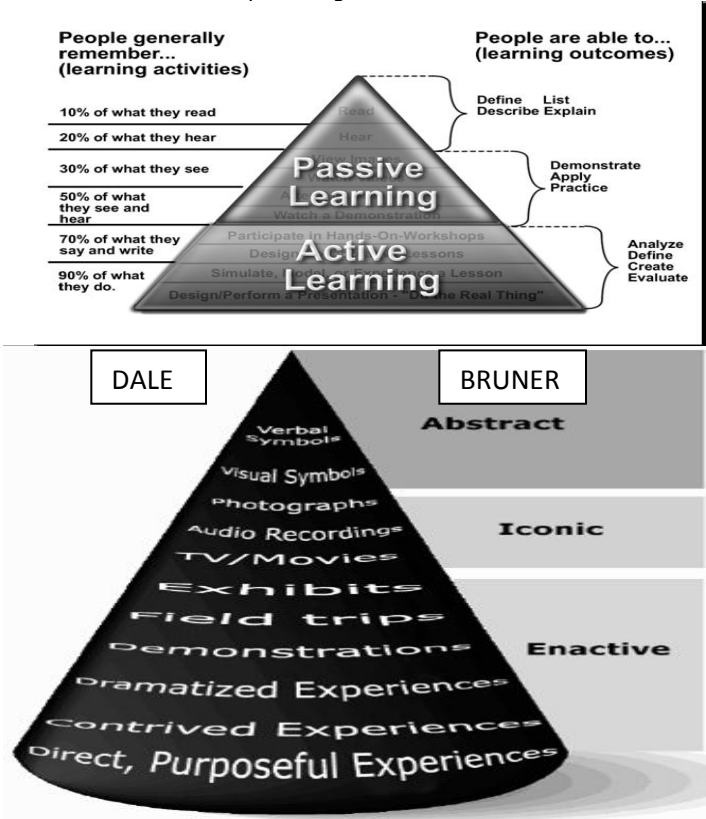
- help get closer to certain realities that are no longer available at first hand
- stirring and attention getting
- participant learns to understand intimately the character he portrays
- teaches cooperative work
- Disadvantages: time consuming without commensurate results
 - participation is limited to few individuals

Contrived Experiences

- an "editing" of reality
- substitutes for confusing or unmanageable first – hand experiences
- easier to handle, manipulate or operate
- Disadvantages:
 - simplification leads to misconceptions, distorted views, and incomplete pictures of reality
 - no freedom to handle expensive or fragile models, mock – ups, specimens, etc.

Direct, Purposeful Experiences

- unabridged version of life itself
- direct participation with responsibility for the outcome
- the basis for the most effective and lasting learning
- Disadvantage:
 - not all things can be learned through direct, first hand experiencing



Variety of Methods

Vary methods used to incorporate all elements of active learning

- Small group work
- Brainstorming
- Creative work
- Games
- Presentations

- Case studies
- Role-plays
- Simulations
- Demonstrations

Peer Teaching

- Build on the wealth of expertise of participants
- Encourage participants to draw on and share their experiences with others
- Allow participants to answer each others' questions (not the facilitator!)

Interactive Process

Use activities that build on and overlap concepts and skills learned to:

- Reinforce learning
- Provide more opportunities to digest and integrate

Real-world Application

- Solicit examples of hypothetical or real problems
- Relate new learning to participant's life situation
- Develop individual action plans for application in the workplace
- When possible, make field visits

The Essential Smart Classroom

" a multimedia room design that features a fixed overhead (video / data) projector and screen, a computer, and built-in VCR or DVD players, audio cassette and CD. What makes them smart is a room control system that enables instructors to easily operate A/V equipment"

Designs that Improve Teaching and Learning

- Empower Faculty
- Emphasize Flexibility
- Encourage Student Interaction
- Stress Simplicity
- Expand Connectivity
- Contain Costs
- Sweat Details

Bloom's Taxonomy for the Digital Age - Integrating Technology

- Knowledge** -Internet, Electronic encyclopedias
- Comprehension**-E-mail, inspiration, overhead
- Application**-Curriculum software
- Analysis** -Spreadsheets, Boolean searchers
- Synthesis**-Multimedia Production
- Evaluation**-Video, Digital cameras production, rubrics

Software Types by their Cognitive Demand

Web sites, CD-ROMs → Knowledge, Comprehension

Drill and practice or other tutorial software → Knowledge, Comprehension, Application

Simulation software → Application, Analysis

Tool software such as spreadsheets, databases, word processing, concept mapping, etc. → Application, Analysis, Synthesis, Evaluation

Copyright Law hhhh

- Copyright, a form of intellectual property law, protects original works of authorship including literary, dramatic, musical, and artistic works, such as poetry, novels, movies, songs, computer software, and architecture. Copyright does not protect facts, ideas, systems, or methods of operation, although it may protect the way these things are expressed.

Word Processing in the Classroom

Word processing can help teachers prepare classroom materials they previously typed or wrote out by hand. These include handouts or other instructional materials, lesson plans and notes, reports, forms, letters to parents or students, flyers, and newsletters.

Spreadsheets in the Classroom

A spreadsheet application software is a computer program that allows the user to enter text and numbers into a table

with rows and columns, and then maintain and manipulate those numbers using the table structure

Electronic spreadsheet programs organize and manipulate numerical data.

Databases in the Classroom

Database software is a computer programs that allow users to store, organize, and manipulate information, including both text and numerical data. On the other hand, database is a system designed to store information in a way that makes it easy to locate later.

Multimedia in the classroom

Multimedia is a combination of text, audio, still images, animation, video and interactive content forms. Multimedia is presented (most of the time) in more than one medium. Using different types of media in the classroom can help to enhance student learning.

Publication in the Classroom

Creating publication in the classroom will strengthen the relationship of the parents and the teacher. Creating newsletters to effectively communicate with parents and patrons is of great help to teachers. The teacher also can create other documents such as banners, posters, business cards, and charts for use in the classroom.

Hypermedia

The name hypertext, or hypermedia, has been applied to networks of nodes (also called articles, documents, files, cards, pages, frames, screens) containing information (in text, graphics, video, sound, and so on) that are connected by links (also called pointers, cross-references, citations).

Email Overview

- Remember, email belongs to the company [Electronic Communications Privacy Act, 1986]
- It is retrievable even after deletion
- Monitoring by the company is legal, that is, you have no expectation of privacy
- If you have a private email address that also gets used for business, expect no privacy here either

Email in Practice

- Don't write anything you wouldn't want to see on the front of the New York Times
- Don't ever respond in anger
- DON'T USE ALL CAPS, IT LOOKS LIKE YOU ARE SHOUTING!!!
- Understand your company's email policy

Teleconferencing--Pros and Cons

- Advantages: shorter meetings, more task oriented, more organized, more equal participation regardless of status, more open exchange of ideas, less likely to be swayed by group norms, less threatening
- Disadvantages: fewer nonverbal cues to aid interpretation, tech uncertainty

Videoconferencing

- Advantages: almost all of teleconferencing plus better chance of getting nonverbals
- Disadvantages: startup costs (\$20-25,000), tech uncertainty

Key Elements for creating 21st century skills

- Emphasize core subjects
- Emphasize learning skills
- Use 21st century tools to develop learning skills
- Teach and learn in the 21st century context
- Teach and learn 21st century content
- Use 21st century assessments that measure 21st century skills

21st Century Learning Skills

- Information and Media Literacy Skills
- Communication Skills
- Critical Thinking and Systems Thinking
- Problem Identification, Formulation and Solution
- Creativity and Intellectual Curiosity
- Interpersonal and Collaborative Skills
- Self-Direction

- Accountability and Adaptability
- Social Responsibility

Traditional vs. Integrated

TRADITIONAL

- Teacher-oriented
- Single-sense stimulation
- Single-path progression
- Single media
- Isolated work
- Information delivery
- Passive learning
- Factual, knowledge-based learning
- Reactive response
- Isolated, artificial content

INTEGRATED

- Student-centered
- Multi-sensory stimulation
- Multi-path progression
- Multi-media
- Collaborative work
- Information exchange
- Active/exploratory/inquiry-based learning
- Critical thinking and informed decision-making
- Proactive/planned action
- Authentic/real-world context

The ARCS Model of Motivational Design

(Attention, Relevance, Confidence, Satisfaction)

The ASSURE Model

- Analyze learners
- State objectives
- Select methods, media, and materials
- Utilize media and materials
- Require learner participation
- Evaluate and revise

How To . . . Conduct a Debriefing: The Four-D Procedure

- | | |
|---------|---|
| Step 1. | Decompressing (Feelings)
"How did you fell when . . . ?" |
| Step 2. | Describing (Facts)
"What happened when you . . . ?"
"What real-life situation does this resemble?" |
| Step 3. | Drawing Comparisons (Transfer)
"How does the scoring system compare with real life costs and payoffs?" |
| Step 4. | Deriving Lessons (Application)
"How can you apply these learning to real life?" |

Objectives Checklist

- | | |
|------------|--|
| Audience | ✓ Specifies learners |
| Behavior | ✓ Learner performance
✓ Observable behavior
✓ Real-World skill |
| Conditions | ✓ Equipment, tools, aids, or references that may be used
✓ Environmental conditions |
| Degree | ✓ States standard for acceptable performance (time, accuracy, proportion, quality) |

Challenges for Teachers Using Technology

- Learning how to use a variety of technology applications;
- Using, adapting, and designing technology-enhanced curricula to meet students needs;
- Expanding content knowledge;
- Taking on new roles; and
- Responding to individual students

Community based learning

- Communities
 - Regional
 - Sectoral
 - Supply Chain based
 - Around prominent individuals or groups
 - Auto-Forming Communities
- Learning within the communities

- Explicit
- Tacit knowledge
- Supported by knowledge platform

Comparison of Distance Education and Online Learning

Distance Education (Chapter 7)	Online Learning (Chapter 8)
Learning via telecommunications media—radio, telephone, television	Instruction delivered by computer, often using networks
Communication is often one way	Communication is usually two way
Similarities = Both can be used as part of organized instructional program.	
Both can be used at all levels (P–20).	
Both can be used in both formal and informal education.	
Both can be used for all subject-matter disciplines.	
Both done with physical separation of learners and teacher.	

The real problem is not whether machines think but whether men do.

-B. F. Skinner Contingencies of Reinforcement, 1969

Prepared by:
Dr. Helen Bihag – BOHOLANO
Associate dean, cte
Cebu Normal University

TAXONOMY OF EDUCATIONAL OBJECTIVES

THINK . . .

- What do you know about the three domains of learning?
- What do you know about Bloom’s Taxonomy?
- Where will WE use this information later?

THE THREE DOMAINS

A. COGNITIVE DOMAIN

- **RATIONAL LEARNING: THINKING-** Emphasis upon knowledge, using the mind, and intellectual abilities.

Bloom’s Taxonomy Provides

1. The basis for creating C-L-E-A-R student learning expectations:

- Comprehensible to students
- Learner-Centered
- Evident, observable in fulfillment
- Attainable, but of a high standard
- Related to the content and goals

1. Knowledge (low level thinking)

The learner should be able to

- Acquire specific facts, ideas, or vocabulary
- Recall and move information from short-term to long-term memory

Cues and starter verbs include

- Define
- List
- Record
- Repeat
- Name
- Recall

2. Comprehension (low level thinking)

Learners should be able to

- Grasp the meaning of material learned
- Communicate what has been learned and interpret it
- Reach understanding

Cues and starter verbs include

- Describe
- Discuss
- Explain
- Identify
- Locate
- Report

3. Application (low level thinking)

The learner should be able to

- Use learned knowledge
 - in new or concrete ways, or
 - to solve new problems

Cues and starter verbs include

- Apply
- Illustrate
- Demonstrate
- Dramatize
- Employ
- Use

4. Analysis (high level thinking)

The learner should be

able to

- Take ideas and knowledge apart
- Dismantle concepts into their components and
 - seek links between concepts (compare)
 - find what is unique (contrast)

Cues and starter verbs include

- Analyze
- Calculate
- Distinguish
- Examine
- Experiment
- Relate
- Solve

5. Synthesis (high level thinking)

Learner should be able to

- Re-organize parts to create a new or original concept or idea
- Make predictions based on analysis of knowledge

Cues and starter verbs include

- Arrange
- Compose
- Formulate
- Construct
- Predict
- Design
- Create

6. Evaluation (high level thinking)

Learner should be able to

- Make judgments or decisions based on logical criteria or conditions
- Rate or assess conclusions
- Make valid choices

Cues and starter verbs include

- Assess
- Select
- Rate
- Estimate
- Compare
- Judge
- Revise
- In my opinion

Bloom’s Taxonomy—Revised

Bloom’s Original Taxonomy	Anderson’s Revised Taxonomy
Knowledge	Remembering
Comprehension	Understanding
Application	Applying
Analysis	Analyzing
Synthesis	Evaluating
Evaluation	Creating

HIGHER ORDER THINKING

Creating

Generating new ideas, products, or ways of viewing things
Designing, constructing, planning, producing, inventing.

Evaluating

Justifying a decision or course of action
Checking, hypothesizing, critiquing, experimenting, judging

Analyzing

Breaking information into parts to explore understandings and relationships

Comparing, organizing, deconstructing, interrogating, finding

Applying

Using information in another familiar situation
Implementing, carrying out, using, executing

Understanding

Explaining ideas or concepts
Interpreting, summarizing, paraphrasing, classifying, explaining

Remembering

Recalling information
Recognizing, listing, describing, retrieving, naming, finding

Remembering: Potential Activities and Products

- Make a story map showing the main events of the story.
- Make a time line of your typical day.
- Make a concept map of the topic.
- Write a list of keywords you know about....
- What characters were in the story?
- Make a chart showing...
- Make an acrostic poem about...
- Recite a poem you have learnt.

Understanding: Potential Activities and Products

- Write in your own words...
- Cut out, or draw pictures to illustrate a particular event in the story.
- Report to the class...
- Illustrate what you think the main idea may have been.

- Make a cartoon strip showing the sequence of events in the story.
- Write and perform a play based on the story.
- Write a brief outline to explain this story to someone else
- Explain why the character solved the problem in this particular way
- Write a summary report of the event.
- Prepare a flow chart to illustrate the sequence of events.
- Make a colouring book.
- Paraphrase this chapter in the book.
- Retell in your own words.
- Outline the main points.

Applying: Potential Activities and Products

- Construct a model to demonstrate how it looks or works
- Practice a play and perform it for the class
- Make a diorama to illustrate an event
- Write a diary entry
- Make a scrapbook about the area of study.
- Prepare invitations for a character’s birthday party
- Make a topographic map
- Take and display a collection of photographs on a particular topic.
- Make up a puzzle or a game about the topic.
- Write an explanation about this topic for others.
- Dress a doll in national costume.
- Make a clay model...
- Paint a mural using the same materials.
- Continue the story...

Analyzing: PotentialActivities and Products

- Use a Venn Diagram to show how two topics are the same and different
- Design a questionnaire to gather information.
- Survey classmates to find out what they think about a particular topic. Analyse the results.
- Make a flow chart to show the critical stages.
- Classify the actions of the characters in the book
- Create a sociogram from the narrative
- Construct a graph to illustrate selected information.
- Make a family tree showing relationships.
- Devise a role-play about the study area.
- Write a biography of a person studied.
- Prepare a report about the area of study.
- Conduct an investigation to produce information to support a view.
- Review a work of art in terms of form, color and texture.
- Draw a graph
- Complete a Decision Making Matrix to help you decide which breakfast cereal to purchase

Evaluating: Potential Activities and Products

- Write a letter to the editor
- Prepare and conduct a debate
- Prepare a list of criteria to judge...
- Write a persuasive speech arguing for/against...
- Make a booklet about five rules you see as important. Convince others.
- Form a panel to discuss viewpoints on....
- Write a letter to. ..advising on changes needed.
- Write a half-yearly report.
- Prepare a case to present your view about...
- Complete a PMI on...
- Evaluate the character’s actions in the story

Creating: Potential Activities and Products

- Use the problem solving strategy to invent a new type of sports shoe
- Invent a machine to do a specific task.
- Design a robot to do your homework.
- Create a new product. Give it a name and plan a marketing campaign.
- Write about your feelings in relation to...
- Write a TV show play, puppet show, role play, song **or** pantomime about..
- Design a new monetary system
- Develop a menu for a new restaurant using a variety of healthy foods
- Design a record, book or magazine cover for...
- Sell an idea
- Devise a way to...
- Make up a new language and use it in an example
- Write a jingle to advertise a new product.

Designing a New Taxonomy of Educational Objectives

“A place holder for my thinking.”
Robert Marzano

A. Marzano’s Taxonomy

Level’s of Practicing

→Retrieval

→Comprehension

→Analysis

→Knowledge Utilization

→Metacognitive System

→Self-system

KNOWLEDGE DOMAINS

→Information

→Mental Procedures

→Psychomotor Procedures

Characteristics of Marzano’s Taxonomy

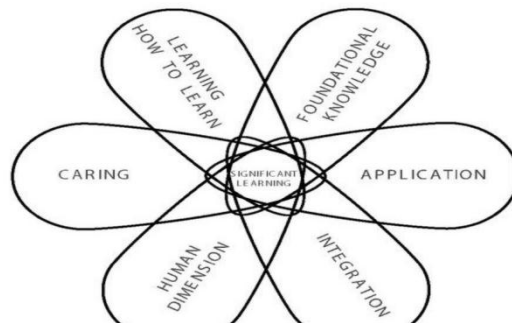
- A “model” or “theory” of human thought (allow for prediction of phenomenon)
- Flow of info ALWAYS (?) (self to metacognitive to cognitive and then knowledge)
- Two dimensions (Three Systems and Knowledge)
- Self-System -interrelated beliefs and goals
- Metacognitive - Goals/learning targets for new tasks
- A “model” or “theory” of human thought (allow for prediction of phenomenon)
- Flow of info ALWAYS (?) (self to metacognitive to cognitive and then knowledge)
- Two dimensions (Three Systems and Knowledge)
- Self-System -interrelated beliefs and goals
- Metacognitive - Goals/learning targets for new tasks

Fink’s Taxonomy of Significant Learning

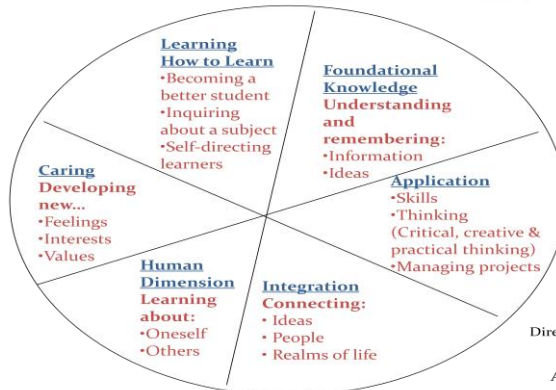
The Taxonomy of Significant Learning

- (1) **Foundational knowledge:** Refers to what is most expected of students; it is the nuts and bolts of the "information of most courses" and is needed for any additional learning about the subject.
- (2) **Application:** Refers to the skills and critical thinking that most professors desire in student learning. This also refers to managing complex projects.
- (3) **Integration:** Integration implies the ability to make connections among different sorts of learned ideas and expands intellectual power.
- (4) **Human dimension:** Deals with the learners' discovering something about themselves and their interactions with others and how this interaction might happen more effectively.
- (5) **Caring:** Caring reflects possibly a change in student attitude either about themselves or what they are learning.
- (6) **Learning how to learn:** Enables a student to continue learning beyond the classroom; i.e. encourages Life Long Learning (Fink, 2003)

THE INTERACTIVE NATURE OF SIGNIFICANT LEARNING



THE TAXONOMY OF SIGNIFICANT LEARNING



Dr. L. Dee Fink
Director, Instructional Development
Program
University of Oklahoma
Author of Creating Significant
Learning Experiences
(Jossey-Bass, 2003)

